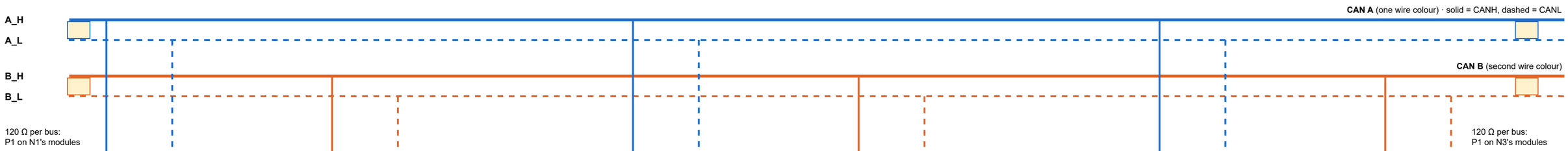
Lab 3 wiring: dual-redundant CAN (CAN A, CAN B), 500 kbit/s, three nodes

Bus order on both buses: N1 (Pi) at one end, N2 in the middle, N3 at the other end. Every module is an XIAO CAN board (MCP2515 + SN65HVD230, 16 MHz crystal, 3.3 V). Each CANH/CANL tap goes through its own jumper wire = the R4.8 fault-injection point. Bus breadboard probe rows A_H, A_L, B_H, B_L. Common GND between Pi, both Nucleos and the bus breadboard.



CAN A module · P1 shorted (bus end)

3V3	← pin 1
GND	← pin 20
D8 SCK	← pin 23 GPIO11
D9 MISO	→ pin 21 GPIO9
D10 MOSI	← pin 19 GPIO10
D7 CS	← pin 24 CE0
D6 INT	→ pin 22 GPIO25

Terminal: CANH, CANL → jumpers ↑

CAN B module · P1 shorted (bus end)

3V3	← pin 17
GND	← pin 25
D8 SCK	← pin 23 GPIO11 (shared)
D9 MISO	→ pin 21 GPIO9 (shared)
D10 MOSI	← pin 19 GPIO10 (shared)
D7 CS	← pin 26 CE1
D6 INT	→ pin 18 GPIO24

Terminal: CANH, CANL → jumpers ↑

SPI + CS + INT + 3V3/GND

N1 cockpit · Raspberry Pi 4
SPI0 (kernel `mcp251x`, 16 MHz osc)
CAN A = can on **spi0.0** (CE0), CAN B = **spi0.1** (CE1)
(can0/can1 names can swap: identify by CE)
Ethernet to laptop proxy (UDP 8000)

Test-point breadboard, N1 (one signal per row)

Row	Signal	Pin
1-2	GND	pin 6 / 9
3	UDP_RX	pin 12 GPIO18
4	CMD_TX	pin 13 GPIO27
5	CAN_TX_A	pin 15 GPIO22
6	CAN_TX_B	pin 16 GPIO23
7	DEG_A (LED)	pin 31 GPIO6
8	DEG_B (LED)	pin 29 GPIO5
9-13	—	

CAN A module · P1 open (middle node)

3V3	← 3V3
GND	← GND
D8 SCK	← PA5 (D13)
D9 MISO	→ PA6 (D12)
D10 MOSI	← PA7 (D11)
D7 CS	← PB6 (D10)
D6 INT	→ PA10 (D2)

Terminal: CANH, CANL → jumpers ↑

CAN B module · P1 open (middle node)

3V3	← 3V3
GND	← GND
D8 SCK	← PA5 (D13) (shared)
D9 MISO	→ PA6 (D12) (shared)
D10 MOSI	← PA7 (D11) (shared)
D7 CS	← PC7 (D9)
D6 INT	→ PB5 (D4)

Terminal: CANH, CANL → jumpers ↑

SPI + CS + INT + 3V3/GND

N2 steering · Nucleo-F401RE
SPI1 at 84/16 = 5.25 MHz, shared by both modules
Zephyr `microchip,mcp2515`, 16 MHz osc
TEST_BTN = on-board B1 (PC13)
IWDG enabled

Test-point breadboard, N2 (one signal per row)

Row	Signal	Pin
1-2	GND	GND
3	CAN_RX_A	PC8 (CN10-2)
4	CAN_RX_B	PC9 (CN10-1)
5	CAN_TX_A	PC10 (CN7-1)
6	CAN_TX_B	PC11 (CN7-2)
7	DEG_A (LED)	PC3 (CN7-37)
8	DEG_B (LED)	PC2 (CN7-35)
9	SRV	PB4 (D5)
10	FL	PB9 (D14)
11	FR	PB8 (D15)
12	RELAY_TX	PB7 (CN7-21)
13	ERR (LED)	PC12 (CN7-3)

CAN A module · P1 shorted (bus end)

3V3	← 3V3
GND	← GND
D8 SCK	← PA5 (D13)
D9 MISO	→ PA6 (D12)
D10 MOSI	← PA7 (D11)
D7 CS	← PB6 (D10)
D6 INT	→ PA10 (D2)

Terminal: CANH, CANL → jumpers ↑

CAN B module · P1 shorted (bus end)

3V3	← 3V3
GND	← GND
D8 SCK	← PA5 (D13) (shared)
D9 MISO	→ PA6 (D12) (shared)
D10 MOSI	← PA7 (D11) (shared)
D7 CS	← PC7 (D9)
D6 INT	→ PB5 (D4)

Terminal: CANH, CANL → jumpers ↑

SPI + CS + INT + 3V3/GND

N3 drivetrain · Nucleo-F401RE
SPI1 at 84/16 = 5.25 MHz, shared by both modules
Zephyr `microchip,mcp2515`, 16 MHz osc
TEST_BTN = on-board B1 (PC13); ERR LED PC12 (CN7-3)
IWDG enabled

Test-point breadboard, N3 (one signal per row)

Row	Signal	Pin
1-2	GND	GND
3	CAN_RX_A	PC8 (CN10-2)
4	CAN_RX_B	PC9 (CN10-1)
5	CAN_TX_A	PC10 (CN7-1)
6	CAN_TX_B	PC11 (CN7-2)
7	DEG_A (LED)	PC3 (CN7-37)
8	DEG_B (LED)	PC2 (CN7-35)
9	PWM_SET	PC6 (CN10-4)
10	PWM_OUT	PA0 (A0)
11	DIR_A	PB3 (D3)
12	RL	PB9 (D14)
13	RR	PB8 (D15)

Common GND: Pi, N2, N3 and the bus breadboard (wired before anything else)

Notes. (1) Module pin D9 is MISO and D10 is MOSI on these boards (the handout figure has them swapped). (2) Termination uses the P1 pad on N1's and N3's modules; P1 sits on the module side of the jumper, so pulling an end node's jumper also removes that bus end's terminator (R4.3 wants it on the bus side; known trade-off, discrete 120 Ω on the rail ends would fix it). (3) CANH–CANL measured powered off: bus A 58 Ω. (4) Bus levels (photos 8–9): idle ≈ 2.2 V both lines, dominant differential 2.07 V (A) / 1.97 V (B).